

PEP Simulation Overview

- The Philippines Energy Plan (PEP) 2023 simulation models the economic and health impacts of energy sector transitions
- We compare a baseline scenario to the PEP 2023 scenario over a 50-year time horizon
- Key dimensions analyzed:
 - Energy sector parametrization (TFP, costs, investments)
 - Environmental outcomes (CO₂e, PM_{2.5})
 - Health impacts (mortality rates, deaths averted)
 - Macroeconomic effects (GDP, capital, consumption, labor)
 - Fiscal implications (government spending, debt)
 - Distributional impacts (prices, income by ability group)

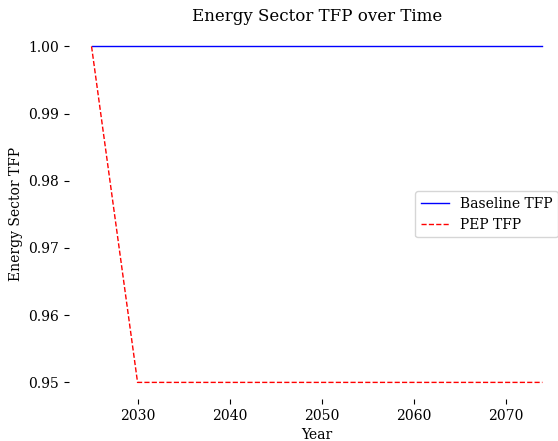
Input-Output Table for Multi-Industry Calibration

- Input-output table used for multi-industry calibration in PEP simulation

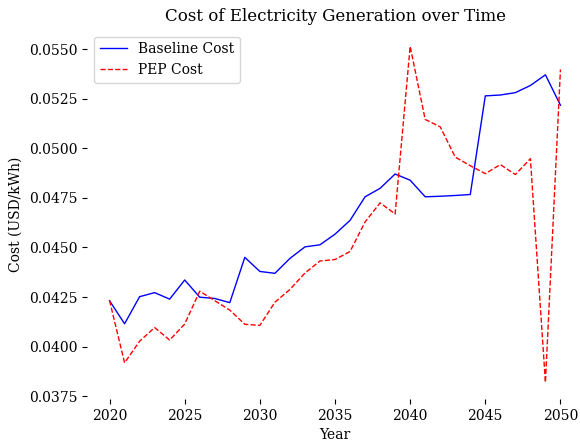
Input-Output Table used for Multi-Industry Calibration in PEP Simulation

	Natural Resources	Electricity	Construction, T
Food	0.120	0.000	
Energy and water	0.021	0.107	
Non-durables	0.043	0.006	
Durables	0.011	0.011	
Services	0.041	0.003	

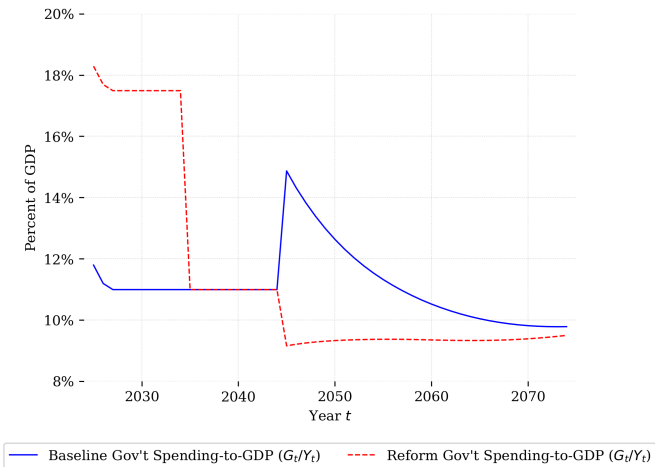
Energy Sector TFP over Time



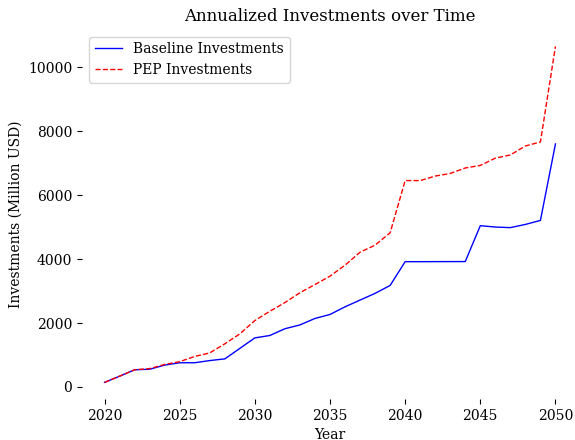
Cost of Electricity Generation over Time



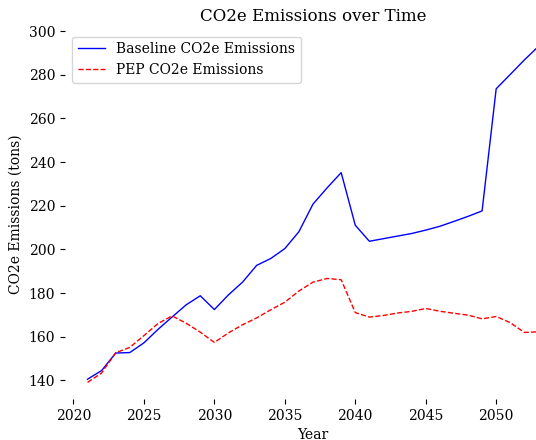
Government Spending over Time



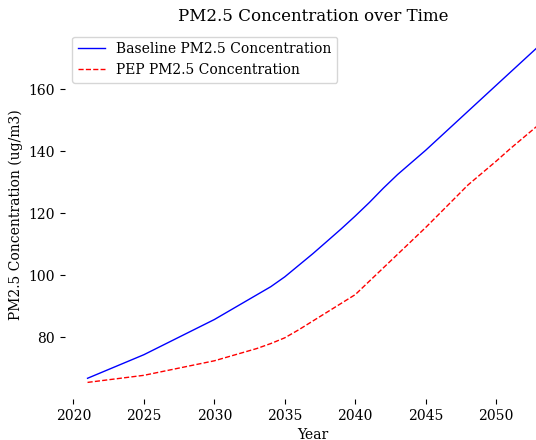
Annualized Investments over Time



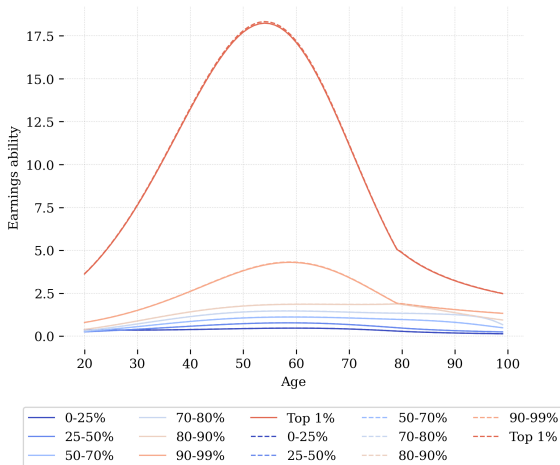
CO₂e Emissions over Time



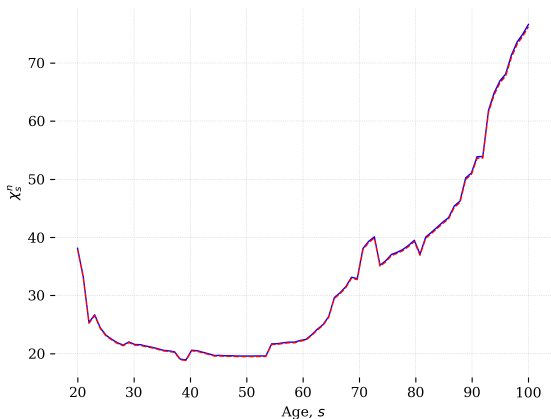
PM2.5 Concentration over Time



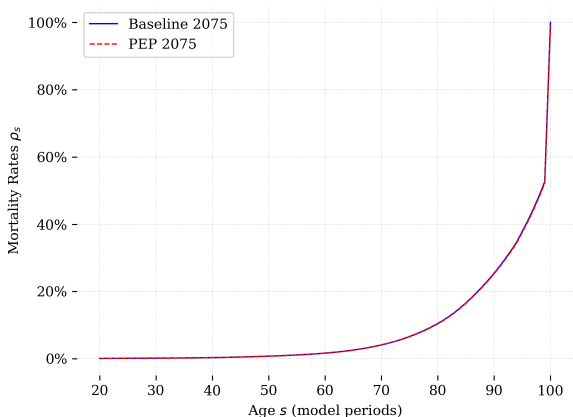
Ability Profiles in Baseline and PEP



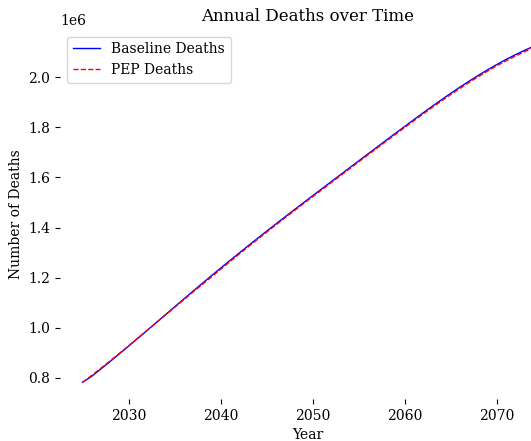
Chi_n Profiles in Baseline and PEP



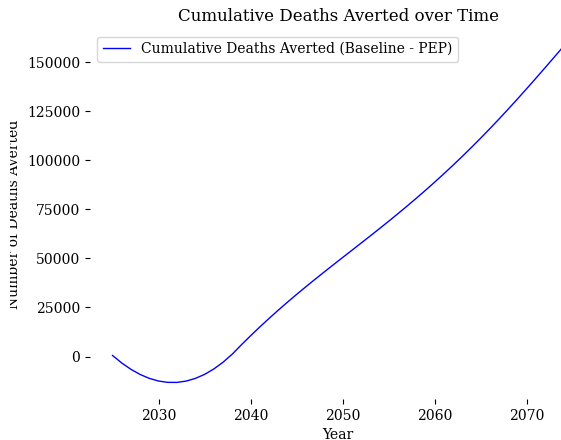
Mortality Rates in Baseline and PEP



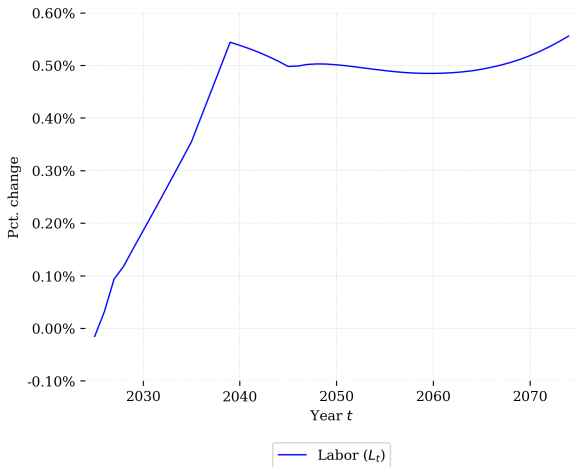
Annual Deaths over Time



Cumulative Deaths Averted over Time

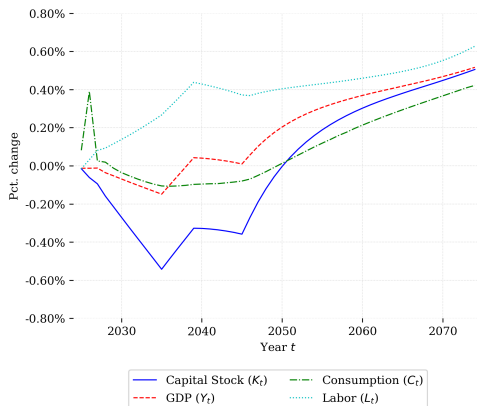


Labor Supply over Time

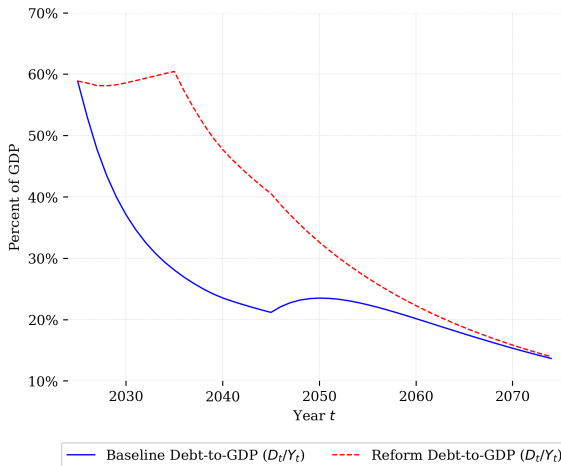


Aggregate Variables over Time

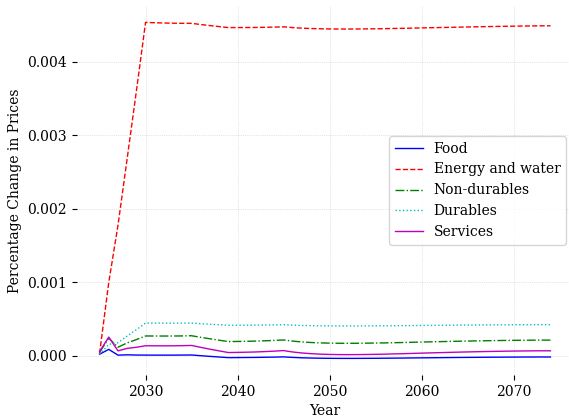
- Percentage changes in capital (K), output (Y), consumption (C), and labor (L)



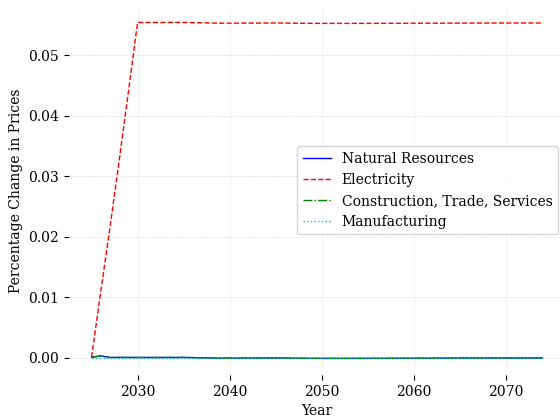
Debt-to-GDP Ratio over Time



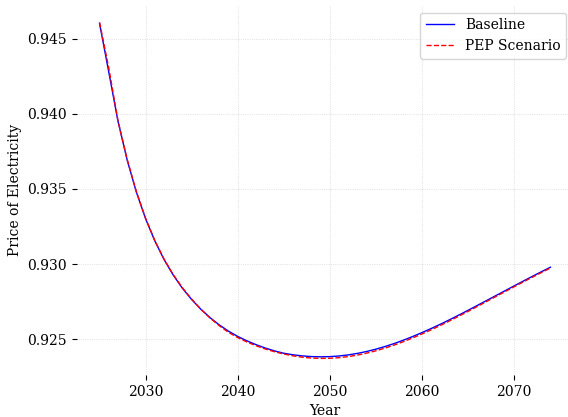
Percentage Change in Consumption Good Prices



Percentage Change in Production Good Prices

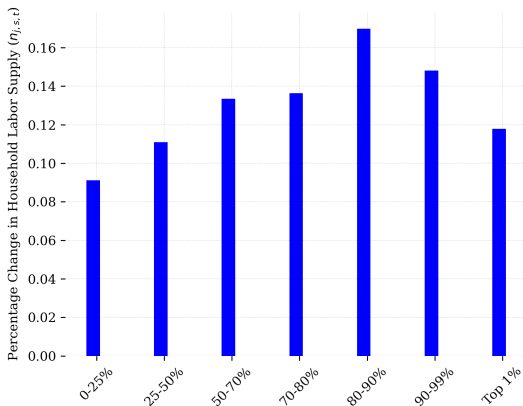


Price of Electricity



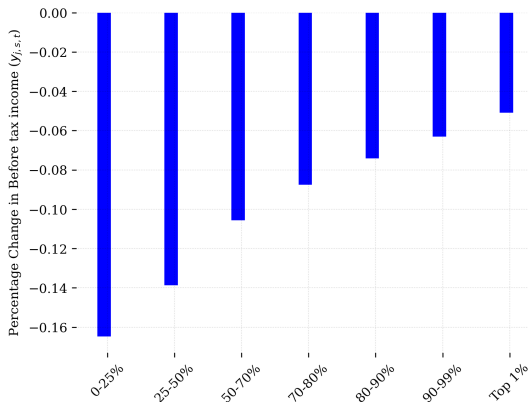
Percentage Change in Labor Supply by Ability Group

- Changes 10-20 years after policy implementation



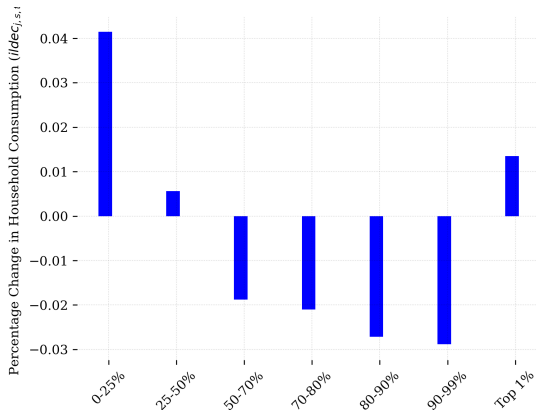
Percentage Change in Income by Ability Group

- Changes 10-20 years after policy implementation



Percentage Change in Consumption by Ability Group

- Changes 10-20 years after policy implementation



Key Findings

- Energy Transition:
 - PEP scenario shows increased energy sector TFP over time
 - Lower electricity generation costs in PEP scenario
 - Significant upfront investments required
- Environmental Benefits:
 - Reduced CO₂e emissions
 - Lower PM_{2.5} concentrations
- Health Impacts:
 - Improved survival rates and lower mortality
 - Substantial cumulative deaths averted
 - Increased labor supply due to better health

Key Findings (continued)

- Macroeconomic Effects:
 - Positive impacts on capital, output, consumption, and labor
- Fiscal Implications:
 - Government spending increases for energy investments
 - Changes in debt-to-GDP ratio
- Distributional Effects:
 - Changes in relative prices across sectors
 - Electricity prices adjust over time
 - Differential impacts across ability groups